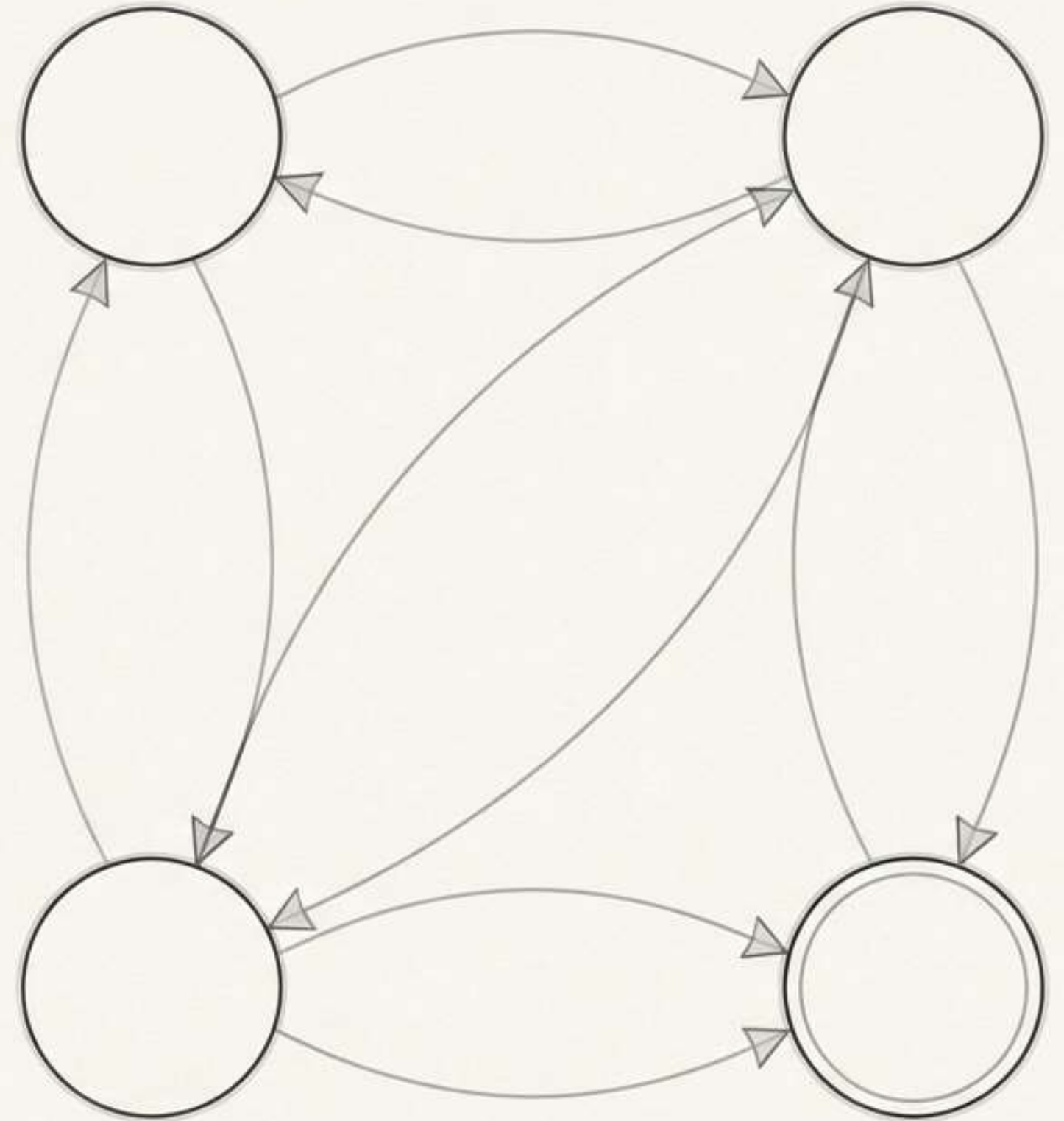


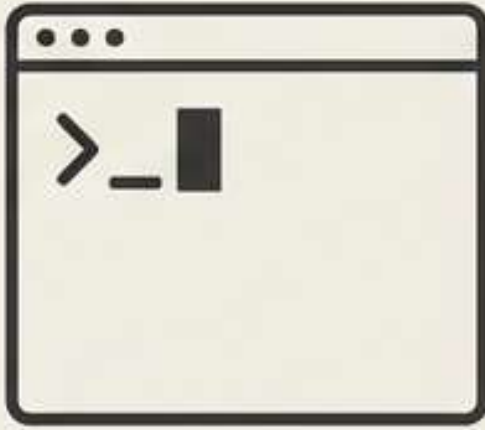
Designing a Deterministic Finite Automaton

A visual guide to constructing a machine that accepts strings of exactly length two.



The Objective & The Constraints

Machine Input



Alphabet (Σ): $\{ 0, 1 \}$

Constraint: The machine processes strictly **binary strings**.

Machine Output



Target: String length must equal **EXACTLY 2**.

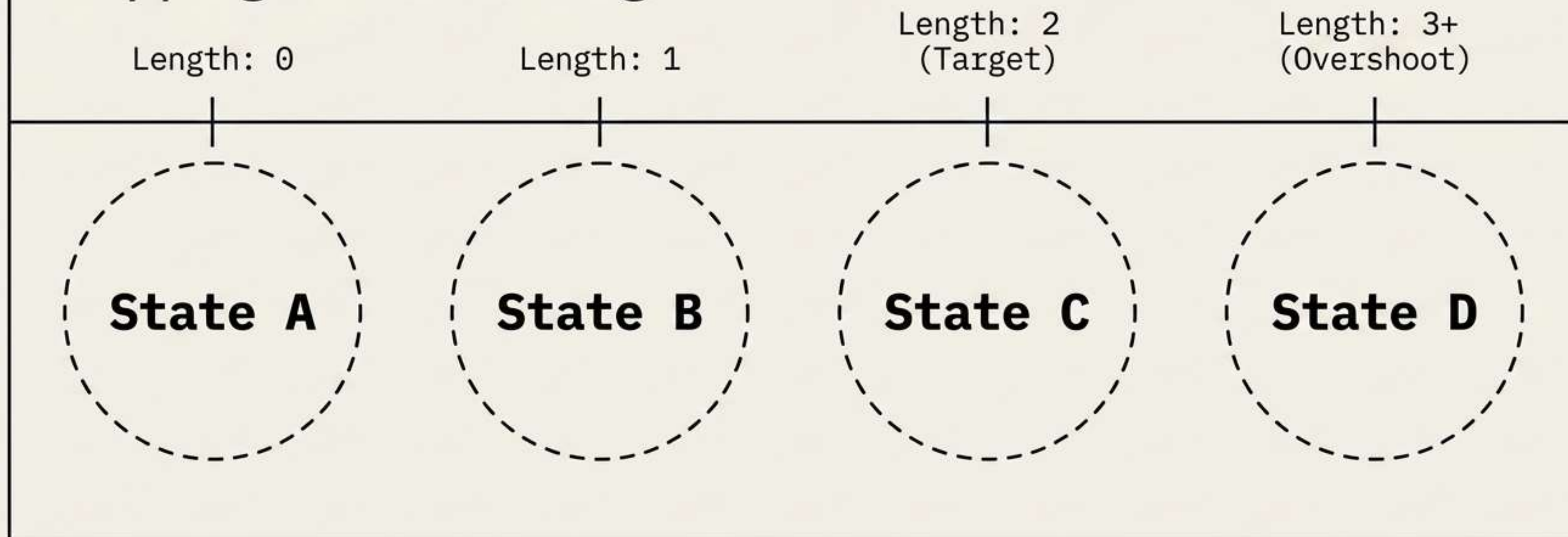
Action: **ACCEPT**  valid strings. **REJECT**  all others.

Defining the Target Language

00	01
10	11

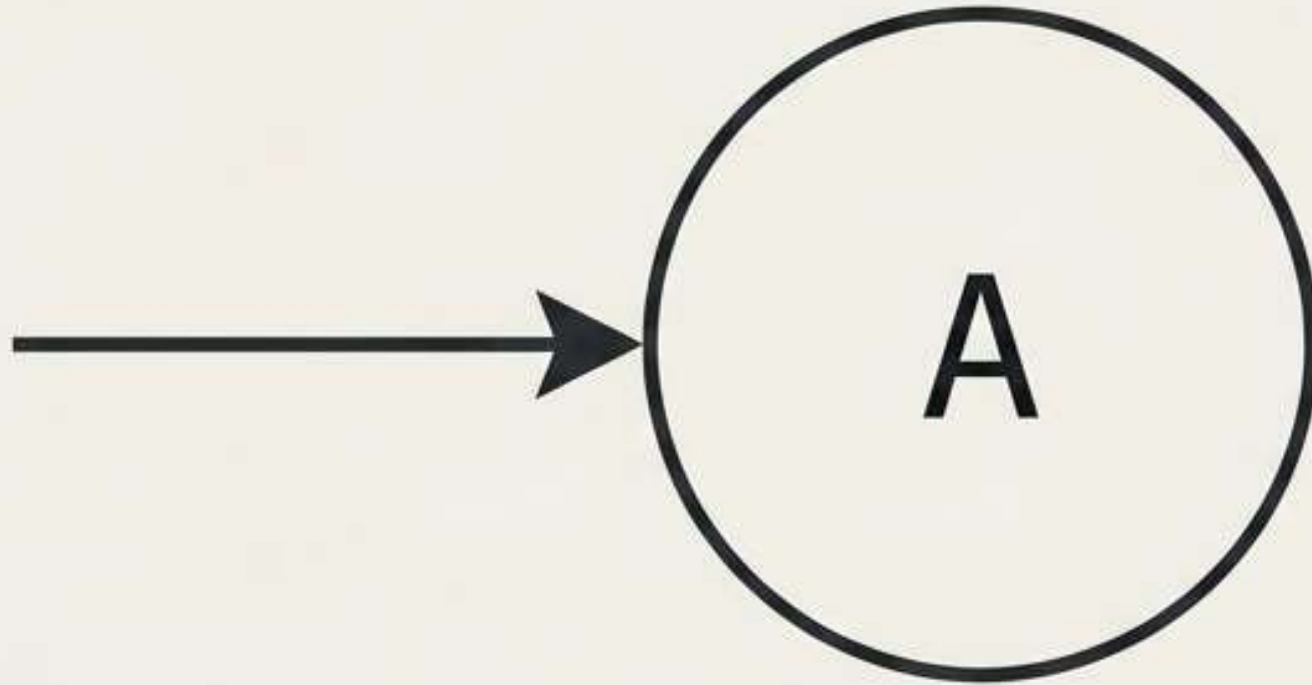
Out of an infinite universe of binary inputs, the DFA must be engineered to isolate and capture strictly these four configurations.

The Conceptual Framework: Mapping States to Length



Key Takeaway: In this architecture, state progression represents character counting. Every input processed advances the machine strictly one step along the timeline.

Step 1: The Point of Origin



System Status

Component:

Initial / Starting State

Mechanism:

The "arrow from nowhere" initializes the machine.

Inputs Processed:

0

Current Length: 0

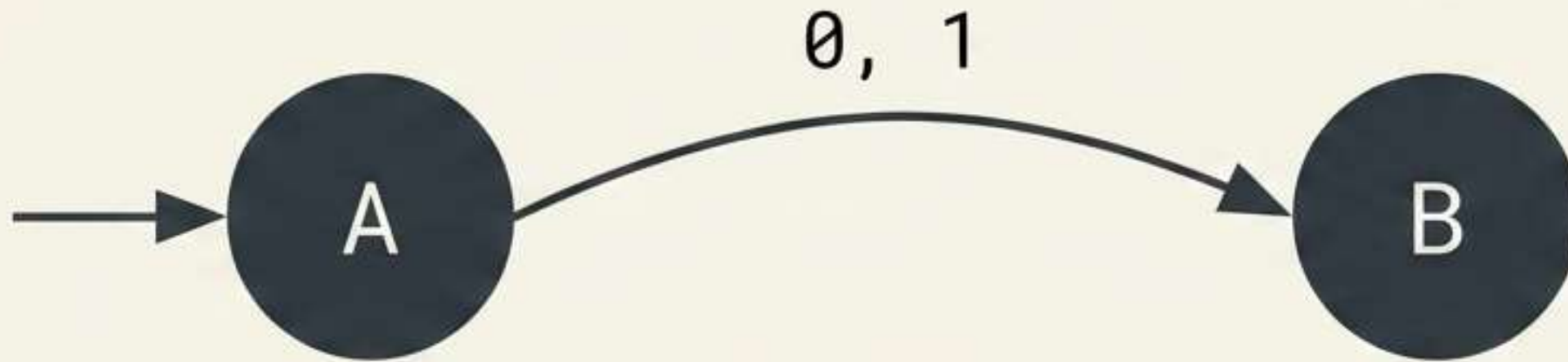
Current Length:

0

Status:

Awaiting input...

Step 2: The First Input



System Status

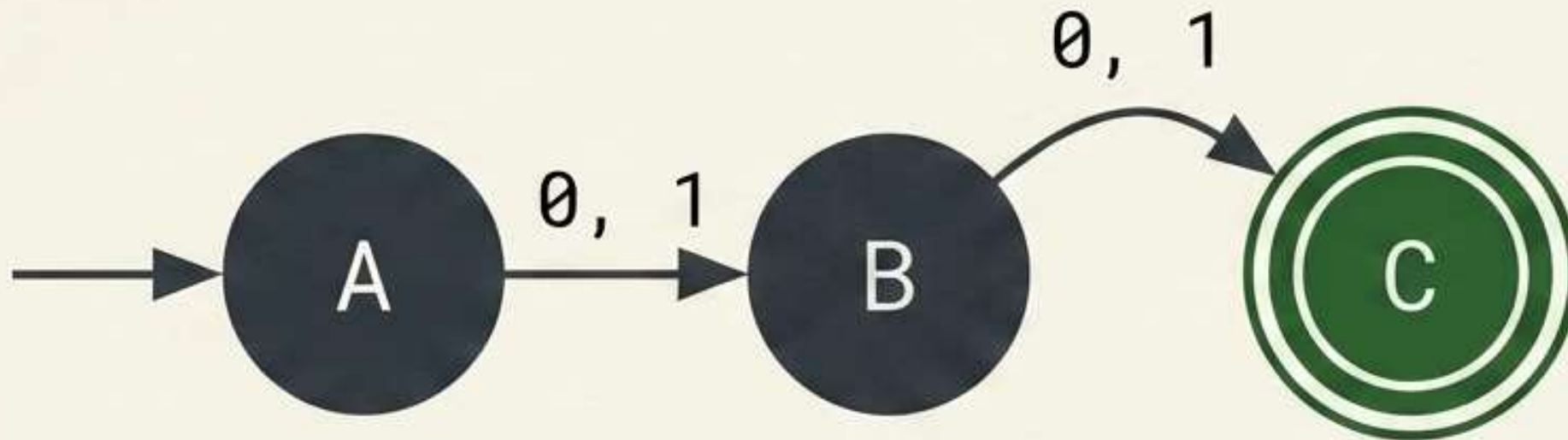
Component: Standard Transition

Mechanism: The receipt of any single binary character advances the machine forward.

Inputs Processed: 1

Current Length: 1

Step 3: Reaching the Target



System Status

Component: Final /
Accepting State

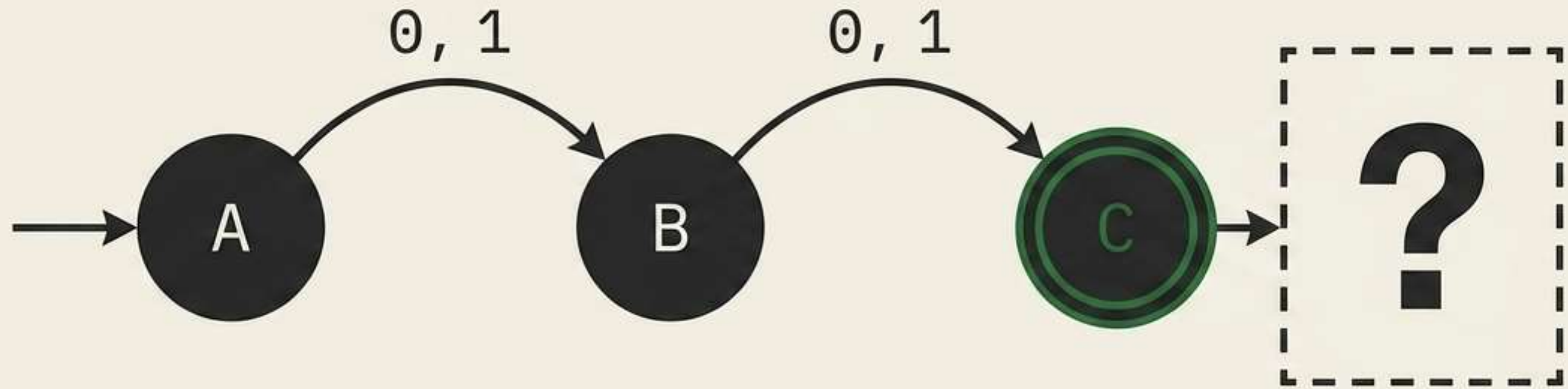
Mechanism: Designated by
a double-circle. Landing
here signifies the
string is valid.

Inputs Processed: 2

Current Length: 2

Status:
Constraint Satisfied.

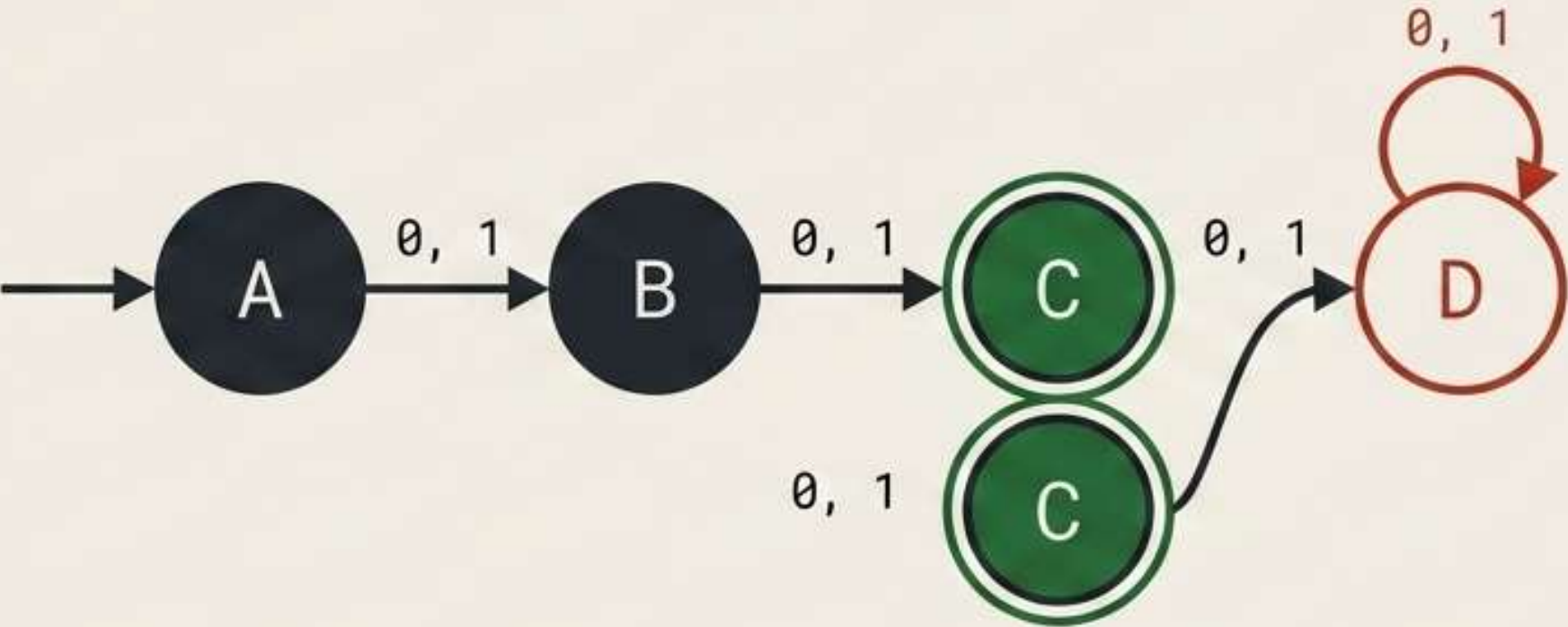
The Boundary Problem



Is the machine complete? NO.

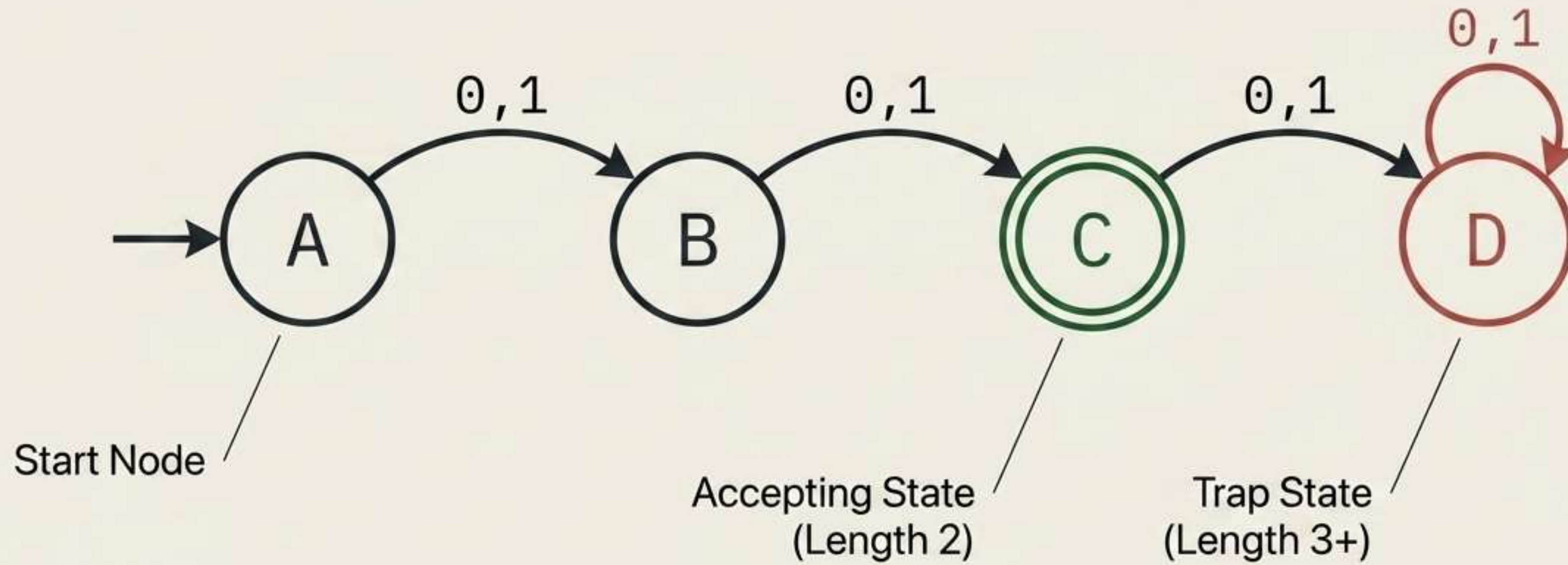
A deterministic machine must map every possible input at every single state.
We have not defined what happens if a third character is introduced.

Step 4: The Trap State

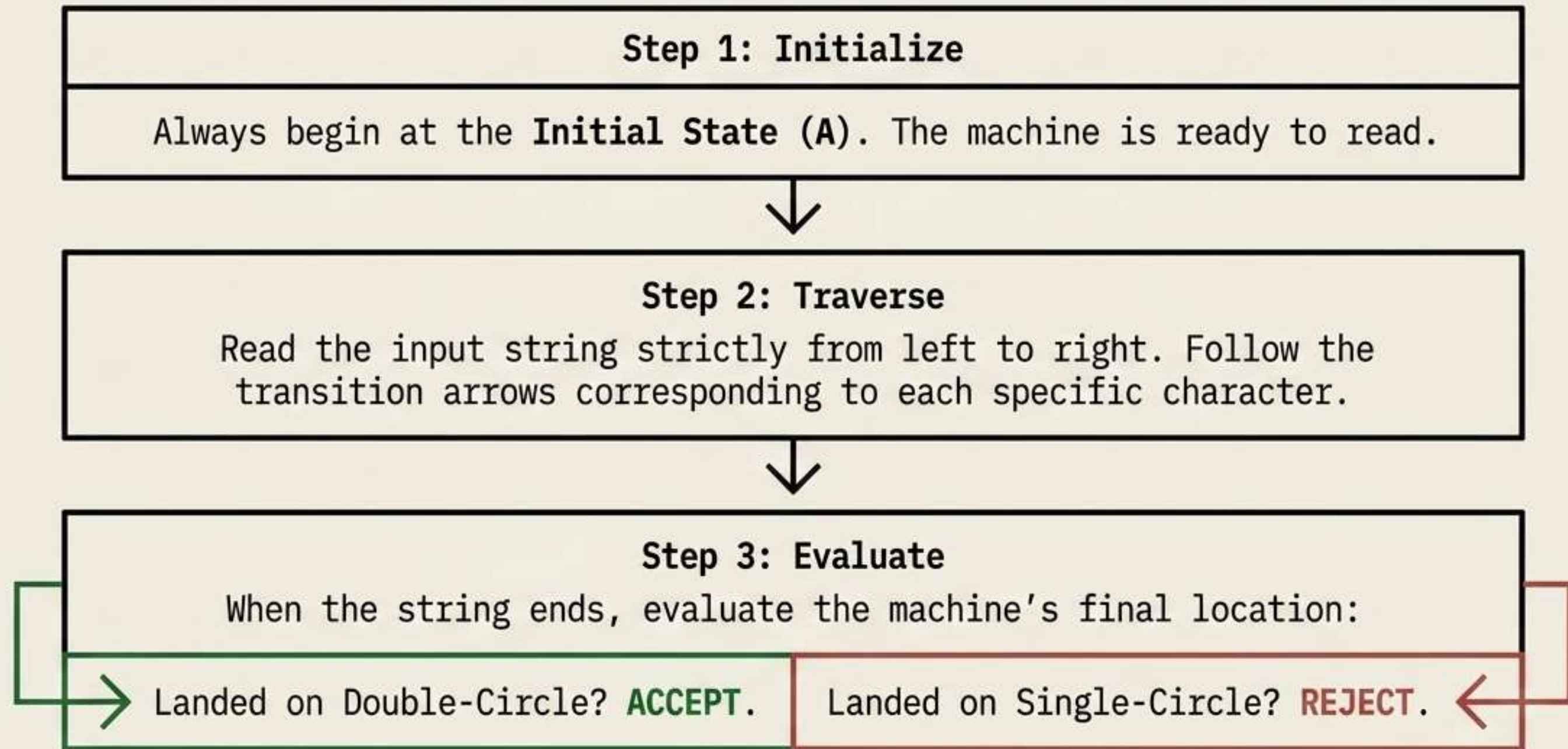


System Status
Component: Dead State / Trap State
Mechanism: Any input received after Length 2 pushes the machine into D. The self-loop creates a permanent rejection zone.
Inputs Processed: 3+
Current Length: 3+
Status: Permanently Rejected.

The Complete Architecture



The Validation Protocol



Test Run 1: Valid Strings

Target String: 00

Start at A.

Input 0 -> Move to B.

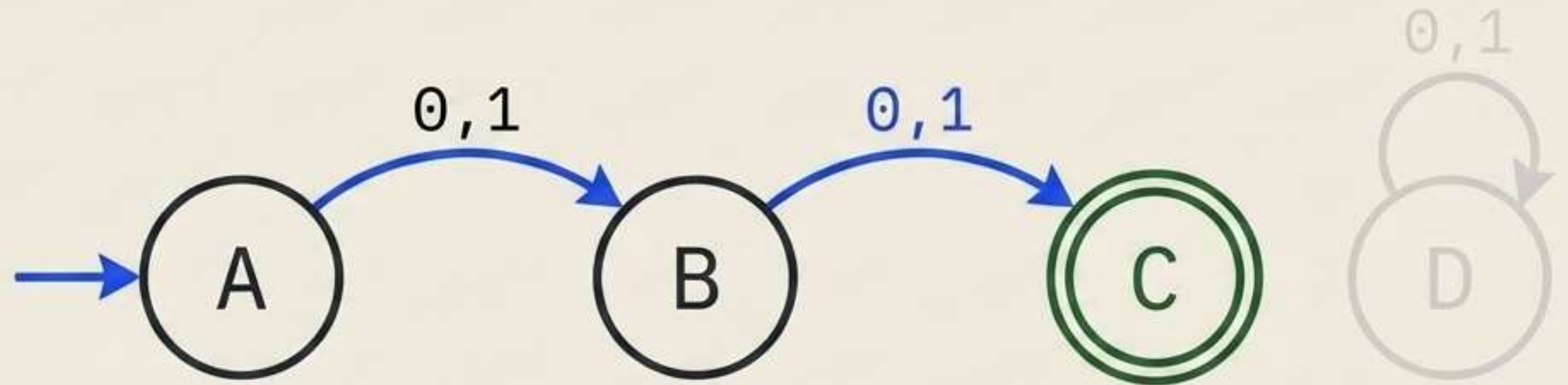
Input 0 -> Move to C.

Input 0 -> Move to C.

String ends.

Result: Landed on Final State C.

[ACCEPTED]



Test Run 2: The Overshoot

Target String: 001

Start at A.

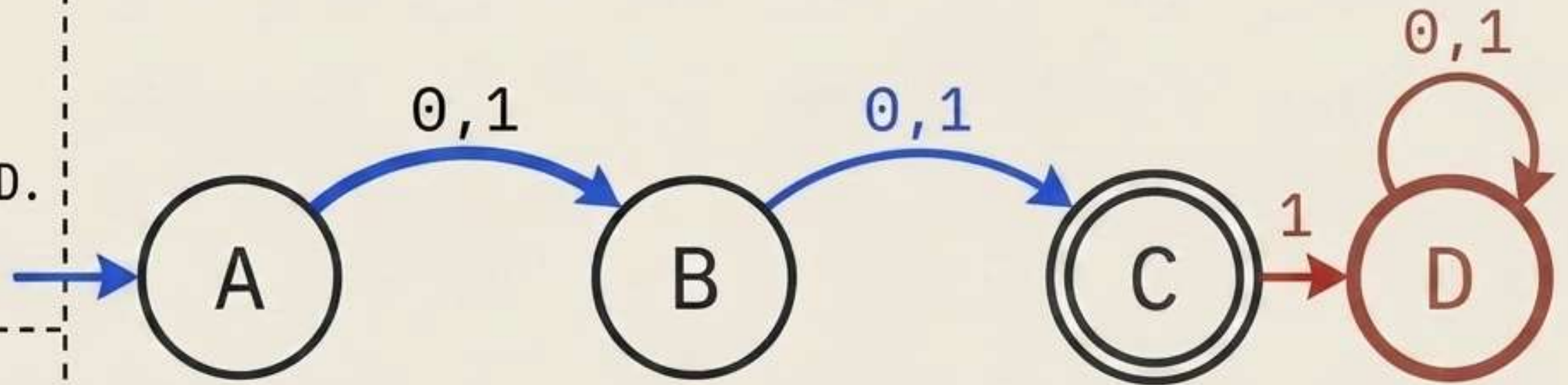
Inputs 00 -> Move to C.

Input 1 -> Forced into D.

String ends.

Result: Landed on Dead State D.

[REJECTED]



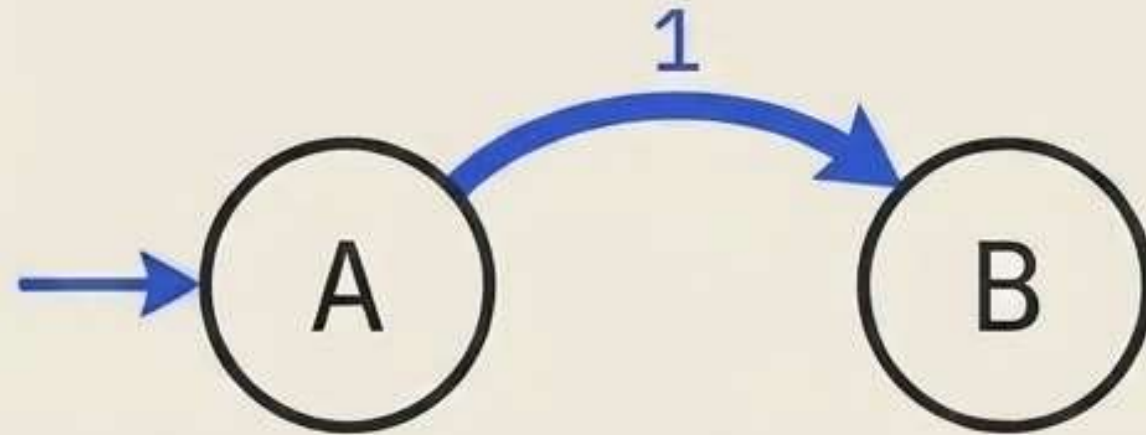
Test Run 3: The Undershoot

Target String: 1

Start at A.

Input 1 -> Move to B.

String ends.



Result: Landed on State B (No double-circle).

[REJECTED]



Synthesis Matrix: Validation Summary

Test String	Input Length	Traversal Path	Final Resting State	Status
00	2	A -> B -> C	State C	[ACCEPT]
10	2	A -> B -> C	State C	[ACCEPT]
001	3	A -> B -> C -> D	State D	[REJECT]
1	1	A -> B	State B	[REJECT]

Conclusion: The architecture flawlessly executes its constraints, isolating strings of exactly length 2 and trapping all other permutations.